

Capstone Project: Forensic Data Lineage in High-Velocity FinTech Systems

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Abstract

This capstone project investigates systemic data corruption points within Tier-1 transactional architectures that are invisible to standard observability and monitoring platforms. The research identifies a pattern, classified as the **Foundational Architectural Flaw (FAF)**, characterized by unlogged, silent data coercion in the ingestion layer. This flaw is demonstrably the root cause of high-cost issues including untraceable model drift, compliance audit exposure, and recurrent engineering debt in production Machine Learning systems. The project concludes with a blueprint for a cryptographic validation layer designed to neutralize FAF at the source.

1. Introduction: The Observability Blind Spot

Modern FinTech architectures rely on robust logging, but a critical failure point exists where data types are coerced without generating audit events. This silent loss of semantic precision (e.g., $\text{FLOAT64} \rightarrow \text{INT32}$) corrupts training data, leading to accelerated Model Drift. Our research proves that the cost of correcting the *consequences* of FAF far exceeds the cost of neutralizing the *cause*.

2. Classification of Foundational Architectural Flaw (FAF)

We classify FAF into three distinct failure modes, with the most pervasive being **FAF 1: Semantic Degradation**. This mode occurs when precision is lost during state transitions, leading to:

- Regulatory Exposure:** Non-compliant data used for reporting.
- Model Decay:** Reduced predictive power in Fraud/AML models.

The flaw persists because monitoring systems only check for pipeline *failure* (a break in the flow), not pipeline *corruption* (a change in semantic value).

3. Solution Methodology: Neutralizing FAF

The proposed solution is a fixed-point, asynchronous validation layer that uses probabilistic and cryptographic hashing to verify data type and value consistency across the ingestion boundary. This layer operates below the observability stack, ensuring data integrity before it reaches the data lake or message queue. The implementation is designed to be completed within a 48-hour fixed-scope diagnostic.

4. Conclusion

The FAF vulnerability is a systematic, solvable problem costing the industry millions in avoidable risk and engineering overhead. Neutralizing FAF is the prerequisite for trustworthy AI governance and compliance readiness.

End of Capstone Summary.